

$$s_1 = \{1, 2, 3\}$$

$$s_2 = \{4, 5, 6\}$$

$$s_1 \cup s_2 = \{1, 2, 3, 4, 5, 6\}$$

$$s_1 \cap s_2 = \emptyset = \{\}$$

intersection

$$s_1.\text{intersection_update}(s_2)$$

$$s_1 \rightarrow \{1, 2, 3\}$$

$$s_2 \rightarrow \{2, 4, 3\}$$

$$s_1 \rightarrow \{2, 3\}$$

$$s_1 = \{1, 2, 3, 4, 5\} \quad s_2 = \{4, 5, 6, 7, 8\}$$

$$s_1 - s_2 = \{1, 2, 3\}$$

$$s_1.\text{difference}(s_2) \rightarrow \{1, 2, 3\}$$

$$s_1.\text{difference_update}(s_2) \rightarrow$$

Symmetric difference

$$s_1 = \{1, 2, 3, 4, 5\} \quad s_2 = \{4, 5, 6, 7, 8\}$$

$$s_1.\text{symmetric_difference}(s_2) = \{1, 2, 3, 6, 7, 8\}$$

Disjoint set, subset, superset

$$s \rightarrow \{1, 2\}$$

$$\text{power set} = 2^n = 2^2 = 4$$

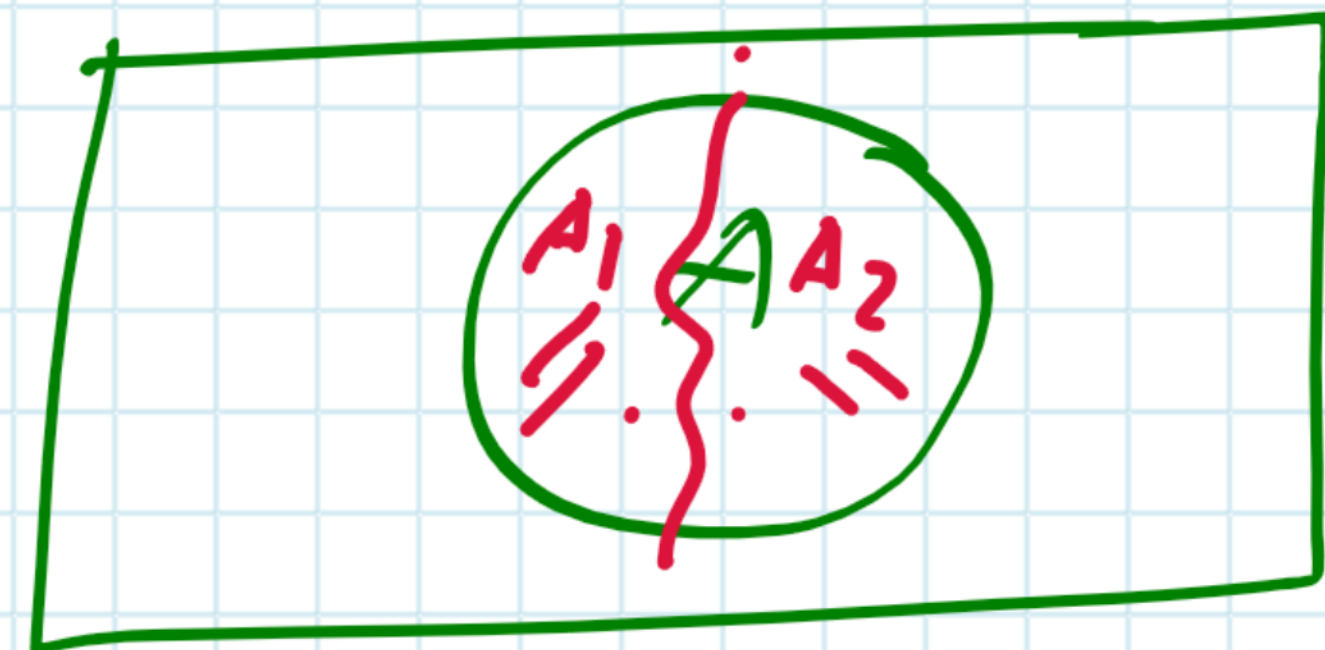
$$\{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

$$s_1 \rightarrow \{1, 2\}$$

$$s_2 = \{4, 5\}$$

$$s_1 \cap s_2 = \emptyset$$

$$s_2 = \{1\}$$



$$A_1 \cap A_2 = \emptyset$$

$$A_1 \cup A_2 = A$$

$$s_1.\text{isdisjoint}(s_2) \rightarrow \text{True/False}$$

$$s_1.\text{issuperset}(s_2) \rightarrow \text{True}$$

$S_1 \rightarrow \{1, 2, 3\}$

power-set $\rightarrow \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\} \}$
0 1 2 3 4 5 6 7

power-set[1] $\rightarrow \{1\}$

power-set[1][0] $\rightarrow 1$

SuperSet

(A) $\rightarrow$ B

$\{1, 2, 3, 4, 5\}$ $\{2, 3\}$

A.issubset(B) $\leftarrow$ True

B.issubset(A) $\leftarrow$ False

A set A is a superset of B if A contains every element of B.

Deep copy vs shallow copy

Deep $S_1 = \{1, 2, 3\}$

$S_2 = S_1$

$S_2 = \{4, 5, 6\}$

print(S2) // {4, 5, 6}

print(S1) // {1, 2, 3}

Shallow

$S_1 = \{1, 2, 3\}$

$S_2 = S_1.copy()$

$S_2 = \{4, 5, 6\}$

print(S2) // {4, 5, 6}

print(S1) // {1, 2, 3}

$S_1 \rightarrow \{1, 2, 3\}$

$S_2 \rightarrow \{4, 5, 6\}$

$[i**2 \text{ for } i \text{ in range}(1, 6) \text{ if } i > 1] \rightarrow [4, 9, 16, 25]$

$\{i**2 \text{ for } i \text{ in range}(1, 6) \text{ if } i > 1\} \rightarrow \{4, 9, 16, 25\}$

Dictionary

$0 \rightarrow 10$

$1 \rightarrow 20$

$2 \rightarrow 30$

'name': print

$L \rightarrow [10, 20, 30]$
0 1 2

$L[0] \rightarrow 10$

key
index

value

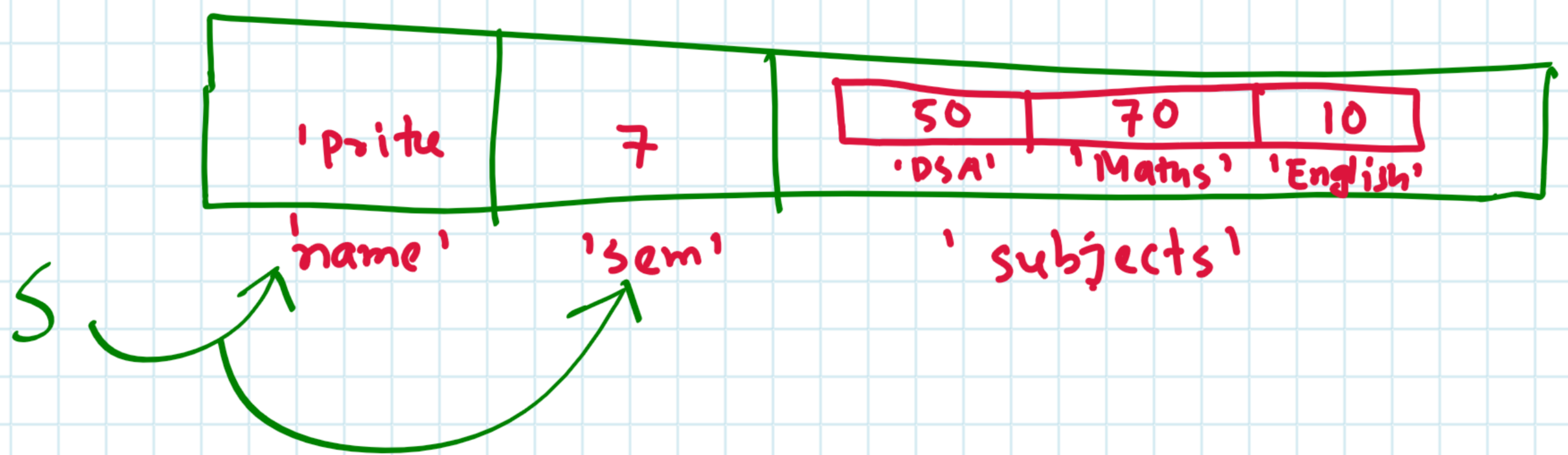
Dictionary

#empty
d = { }

- (i) create
- (ii) Access
- (iii) Adding Key-value pairs
- (iv) Remove " - " "
- (v) Edit
- (vi) operations
 - membership
 - Iteration
- (vii) function
 - len
 - sorted
 - max
 - items
 - keys
 - values
- (viii) Dictionary comprehension

d.items() → dict_items[('name', 'pritu'), ('age', 24)]
 d.keys() → dict_keys['name', 'age', 'gender']
 d.values() → dict_values['pritu', 24, 'Male']

```
s={
    'name':'Pritu',
    'sem':7,
    'subjects':{
        'DSA':50,
        'Maths':70,
        'English':10
    }
}
```



s['name'] ⇒ 'Pritu'
 s['subjects'] ⇒ { 'DSA': 50, 'Maths': 70, 'English': 10 }

s['subjects']['DSA'] ⇒ 50

d1 = { 1:2, 3:4, 4:5 }
 d2 = { 4:7, 6:8 }
 d1.update(d2) →
 d1 = { 1:2, 3:4, 4:7, 6:8 }

Dictionary Comprehension

$\{ \text{key: value for var in iterable} \}$

Set $\leftarrow \{ i ** 2 \text{ for } i \text{ in range}(1, 11) \}$

Dictionary $\leftarrow \{ i : i ** 2 \text{ for } i \text{ in range}(1, 11) \}$

$\{ 1:1, 2:4, 3:9, 4:16, \dots \}$